

ATM DEMO PROGRAM

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An automated teller machine (ATM), also known as automated banking machine or cash machine, is a computerised telecommunication device that provides the clients of a financial institution access to financial transactions in a public space without the need for a cashier, clerk or other bank executive.

Using an ATM, customers can access their bank accounts for cash withdrawals, credit-card cash advances or checking their account balances. ATMs can be found near or inside the premises of banks and also in such locations as shopping malls, airports, grocery stores, petrol/gas stations and restaurants.

Here is a program that demonstrates the working of a typical ATM machine.

ATM basics

Fig. 1 shows the block diagram of an ATM machine. Typically, an ATM machine is made up of:

1. Central processing unit (CPU), to control the user interface and transaction devices
2. Magnetic and/or chip card, to identify the customer
3. PIN pad terminal, often manufactured as part of a secure enclosure
4. Secure cryptoprocessor, generally within a secure enclosure
5. Display – a user-interface device for performing the transaction
6. Function key buttons or touch-screen (usually close to the display), to select various transaction functions
7. Record printer, to provide the customer with details of the transaction
8. Vault, to store parts of the machinery like deposit and dispensing mechanisms, locks and sensors

9. Housing

The customer is identified when he inserts in the ATM machine a plastic ATM card with a magnetic stripe or a plastic smart card. The card contains a unique card number and some security information such as expiry date and card verification value code or card validation code. Authentication is provided when the customer enters the

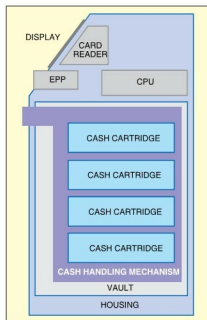


Fig. 1: Block diagram of typical ATM machine



Fig. 2: Program output of ATM demo program



correct personal identification number (PIN).

How the program works

This program (atm.cpp) demonstrates the operations being performed in the machine which are externally visible. Fig. 2 shows the screenshot of the program output. Here "Welcome to ATM Machine" is the message that appears at display area in the ATM. Just below it is the input keys terminal to enter the customer code and other information. The magnetic card is to be inserted in 'Insert Card' slot in the actual ATM machine. The printed detail of the transaction is obtained from the outlet marked 'Receipt'. The user can collect his cash from the 'Cash' slot in the ATM machine.

In this demo program, you don't have to feed the input, as it displays the series of actions involved while accessing an ATM machine.

To run the program in DOS or Windows XP platform, you need the executable file. For this, first compile the atm.cpp program in Turbo C++ IDE and generate the atm.exe file. Before clicking the atm.exe file, make

sure that egavga.bgi, herc.bgi and lcom.chr graphics files are in the same folder as the atm.exe file. In the end, press 'Enter' key to exit the program.

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on the www.efymag.com website.